MANDIMITRA

AI-Powered Agricultural Intelligence
for Maharashtra Farmers

Final Year Diploma Capstone Project
Course Code: 316004

Submitted by: Amar Dighe
Guide: [Faculty Name]

Department of Computer Engineering
[College Name], Maharashtra

Academic Year 2025-2026

Version 2.1.0 | MIT License | March 2026

1. Abstract

MANDIMITRA is a production-grade, AI-powered agricultural decision support system built for Maharashtra farmers. It transforms 3.5M+ government mandi records, satellite weather data, and crop disease imagery into actionable intelligence through three trained ML models -- Crop Risk Advisor (LightGBM, 87.4% accuracy), Price Intelligence Engine (LightGBM, R2=0.933), and Crop Disease Detector (MobileNetV2 CNN, 93.47% accuracy). The system features a FastAPI backend, Next.js 14 frontend, MongoDB database, and a Capacitor-based Android APK with Marathi-first UI. Key capabilities include real-time crop risk assessment, multi-horizon price forecasting with conformal prediction intervals, on-device crop disease detection with treatment recommendations, live mandi price tracking, veterinary emergency SOS with geo-based auto-assignment, and a buyer marketplace. Deployed on Vercel (frontend) and AWS EC2 (backend) with Docker containerization.

2. Introduction

2.1 Problem Statement

Maharashtra farmers face critical challenges: volatile commodity prices causing financial losses, crop diseases going undetected until severe damage, lack of accessible weather-based risk assessment, and limited access to veterinary emergency services in rural areas. Existing solutions are fragmented, English-only, and inaccessible to the average Marathi-speaking farmer.

2.2 Proposed Solution

MANDIMITRA provides an integrated, Marathi-first platform combining three ML models with real-time data from government APIs. Farmers can assess crop risk before sowing, forecast commodity prices with confidence intervals, detect crop diseases via phone camera, track live mandi prices, and request emergency veterinary help -- all in one application.

2.3 Objectives

- Build a data pipeline ingesting 3.5M+ mandi records from AGMARKNET + 10 years weather from NASA POWER
- Train 3 ML models: Crop Risk (LightGBM), Price Forecast (LightGBM), Disease Detection (CNN)
- Develop a FastAPI REST backend with 40+ endpoints and JWT authentication
- Create a responsive Next.js web app + Android APK with Marathi-first UI
- Implement veterinary emergency SOS with geo-based doctor auto-assignment
- Deploy with Docker on AWS EC2 (backend) and Vercel (frontend)

3. Literature Review & Research Papers Implemented

- **MobileNetV2 (Sandler et al., 2018):** Inverted residuals with linear bottlenecks for efficient mobile CNN. Used as backbone for 17-class crop disease detection. ~14 MB model size.
- **MT-CYP-Net (arXiv:2505.12069):** Multi-task crop yield prediction with focal-loss class weighting. Adopted focal-loss weights {Low:1, Medium:10, High:50} for crop risk model, improving High Risk Recall by +3.1%.
- **NeuralCrop (arXiv:2512.20177):** Physics-informed neural features for crop modeling. Implemented GDD (Growing Degree Days), VPD (Vapor Pressure Deficit), and Drought Stress Index as engineered features.
- **Conformal Prediction (Shafer & Vovk, 2005):** Distribution-free prediction intervals. Applied to price forecasting: 80% CI, 90% CI, and 95% CI bands calibrated on held-out data.
- **LightGBM (Ke et al., 2017):** Gradient boosting with leaf-wise growth and histogram binning. Used for both crop risk classification and multi-horizon price regression.

4. System Architecture

4.1 Three-Tier Architecture

Data Layer: AGMARKNET API (3.5M+ records), NASA POWER (10yr weather), Open-Meteo (16-day forecast), MongoDB Atlas (users, bookings, emergencies, buyer prices).

ML/API Layer: FastAPI v2.1.0 on Docker (Python 3.10), LightGBM models (~24 MB), on-device TF.js CNN (saves ~400MB on server), JWT auth, Gemini AI integration.

Presentation Layer: Next.js 14 (Vercel), Capacitor Android APK, Marathi-first bilingual UI, Leaflet maps, Recharts visualization.

4.2 Deployment Architecture

Frontend: Next.js on Vercel (auto-deploy on git push) with API rewrites to backend.
Backend: FastAPI in Docker container on AWS EC2 (t3.micro), port 8000.
Database: MongoDB Atlas (cloud). External APIs: Data.gov.in, Open-Meteo, Gemini AI.
Mobile: Capacitor APK loads UI from Vercel URL -> auto-updates without re-installing APK.

5. Technology Stack

5.1 Frontend

Technology	Version	Purpose
Next.js	14.2.0	React framework (SSR + static)
React + TypeScript	18.2 / 5.3	UI + type safety
Tailwind CSS	3.4.1	Utility-first styling
TensorFlow.js	4.22.0	On-device crop disease CNN
Capacitor	8.1.0	Native Android APK
Framer Motion	11.0.8	Animations
Recharts	2.12.2	Data visualization
Leaflet	1.9.4	Interactive maps
Radix UI	Multiple	Accessible UI primitives

5.2 Backend

Technology	Version	Purpose
Python	3.10+	Core language
FastAPI	0.104+	REST API framework
LightGBM	4.0+	Crop Risk + Price models
scikit-learn	1.3+	ML preprocessing
MongoDB (Motor)	3.3+	Async database driver
PyJWT + bcrypt	2.8+ / 4.0+	Auth & password hashing
Gemini AI (genai)	1.0+	AI treatment advice
Docker	Python 3.10-slim	Containerization

6. Machine Learning Models

6.1 Model 1: Crop Risk Advisor

- **Algorithm:** LightGBM Classifier (gradient boosting)
- **Training Data:** 3.5M+ mandi records + 10yr weather data (36 districts)
- **Classes:** Low Risk, Medium Risk, High Risk
- **Accuracy:** 87.4% (CV: 88.0%)
- **F1 Macro:** 61.08% (CV: 62.02%)
- **High Risk Recall:** 45.89% (+3.1% via focal-loss weights)
- **Key Features:** GDD, VPD, Drought Stress Index, district, temp_std, stage_progress
- **Physics Features:** From NeuralCrop paper -- biology-informed feature engineering
- **Crops Supported:** Soybean, Cotton, Jowar, Bajra, Wheat, Gram, Groundnut, Tur, Maize, Rice, Sugarcane, Onion

6.2 Model 2: Price Intelligence Engine

- **Algorithm:** Multi-horizon LightGBM Regressor (5 separate models)
- **Horizons:** 1-day, 3-day, 7-day, 14-day, 15-day
- **Features:** 60 features (lag, rolling stats, momentum, volatility, calendar, weather)
- **Training Data:** 3.29M records per horizon
- **Prediction Intervals:** Conformal Prediction at 80%, 90%, 95% CI

Horizon	MAE	RMSE	R2	MAPE
1-day	361.72	687.77	0.933	21.3%
3-day	435.38	803.34	0.910	26.2%
7-day	501.82	896.52	0.890	34.6%
14-day	561.81	983.83	0.868	47.6%
15-day	546.28	924.92	0.882	44.8%

6.3 Model 3: Crop Disease Detector

- **Architecture:** MobileNetV2 + custom classification head (transfer learning)
- **Input:** 224x224 RGB images
- **Classes:** 17 (5 crops x diseases + healthy states)
- **Validation Accuracy:** 93.47%
- **Phase 1 (frozen):** 90.35% accuracy
- **Phase 2 (fine-tuned):** 93.47% accuracy
- **Model Size:** ~14 MB (.keras format)
- **Deployment:** On-device via TensorFlow.js (browser/APK) -- not on server
- **Dataset:** 13,324 images across 17 classes

17 Classes: Corn (Common Rust, Gray Leaf Spot, Healthy, Northern Leaf Blight), Potato (Early Blight, Healthy, Late Blight), Rice (Brown Spot, Healthy, Leaf Blast, Neck Blast), Sugarcane (Bacterial Blight, Healthy, Red Rot), Wheat (Brown Rust, Healthy, Yellow Rust)

7. Features & Modules

7.1 Crop Risk Assessment

Farmer selects crop, district, and sowing date. System fetches weather forecast and runs the LightGBM model to predict risk level (Low/Medium/High) with confidence percentage.

7.2 Price Forecasting

Select commodity and market. System runs 5 LightGBM models for 1-day to 15-day price predictions with conformal 80/90/95% confidence intervals displayed on interactive charts.

7.3 Crop Disease Detection

Upload crop leaf photo. MobileNetV2 CNN runs on-device (TF.js) for instant prediction. Cloud fallback uses Gemini AI for detailed treatment advice. Knowledge base provides medicines, fungicides with dosages, and soil mineral deficiency analysis -- all in Marathi.

7.4 Live Mandi Prices

Real-time prices from AGMARKNET (Data.gov.in) with 24-hour auto-refresh. Filter by commodity, market, district. Displays min/max/modal prices.

7.5 Weather Forecast

16-day forecast via Open-Meteo API for all 36 Maharashtra districts. Interactive Leaflet maps with temperature, rainfall, humidity data.

7.6 Veterinary Emergency SOS

One-tap emergency request. System auto-assigns nearest verified vet doctor based on GPS coordinates. Doctor can accept/reject (escalates to next nearest). Full booking management with appointment scheduling.

7.7 Buyer Marketplace

Buyers post daily crop prices. Farmers search by market and crop to find best prices. 14 crops supported with Marathi names. Auto-expiry of stale listings.

7.8 Role-Based Access

Role	Capabilities
Farmer	Risk assessment, prices, disease scan, weather, vet booking, SOS
Doctor	Profile, emergencies, bookings, document upload, GPS update
Buyer	Post prices, manage listings, market selection
Admin	2-step login, verify doctors, dashboard stats

8. API Endpoints (40+)

Authentication (6 endpoints)

Method	Endpoint	Description
POST	/api/auth/signup	Register (farmer/doctor/buyer)
POST	/api/auth/login	Login with JWT
POST	/api/auth/admin/verify	Admin step 1: verify ID
POST	/api/auth/admin/login	Admin step 2: credentials
GET	/api/auth/me	Current user from token
POST	/api/auth/logout	Logout

ML Prediction (6 endpoints)

Method	Endpoint	Description
POST	/api/crop-risk/assess	Crop risk LightGBM prediction
GET	/api/crop-risk/crops	List supported crops
GET	/api/crop-risk/districts	List districts
POST	/api/price/forecast	Multi-horizon price forecast
GET	/api/price/commodities	Top 30 commodities
GET	/api/price/markets	Available markets

Crop Disease (3 endpoints)

Method	Endpoint	Description
POST	/api/crop-disease/analyze	Image upload -> CNN + Gemini advice
GET	/api/crop-disease/classes	List 17 disease classes
GET	/api/crop-disease/treatment/{cls}	Treatment knowledge base

Weather & Market (5 endpoints)

Method	Endpoint	Description
GET	/api/weather/forecast	16-day forecast (Open-Meteo)
GET	/api/weather/districts	Districts with coordinates
GET	/api/market/prices	Live mandi prices
GET	/api/market/commodities	Commodities list
GET	/api/market/summary	Market statistics

Veterinary (14 endpoints) & Buyer (6 endpoints)

Vet module: admin doctor verification, doctor profile/location/bookings/emergencies, farmer booking/SOS creation. Buyer module: CRUD prices, market/crop lists, farmer search.

9. Data Sources

Source	Data	Volume
AGMARKNET (Data.gov.in)	Daily mandi commodity prices	3.5M+ records
Kaggle Historical	Multi-year mandi price archive	Large dataset
NASA POWER	10yr daily weather (precip, temp, humidity)	36 districts
Open-Meteo	16-day weather forecasts	Real-time API
Disease Knowledge Base	Treatment info (IRRI, ICAR, FAO)	17 classes
Crop Lifecycle Config	Growth stages, sowing windows, thresholds	12+ crops

10. Key Statistics

Metric	Value
Total mandi records processed	3.5M+
Maharashtra districts	36
Maharashtra mandis/markets	300+
Disease detection classes	17
Disease detection accuracy	93.47%
Crop risk accuracy	87.4%
Price model R2 (1-day)	0.933
Price model R2 (7-day)	0.890
Forecast horizons	5 (1, 3, 7, 14, 15 days)
ML features (price model)	60
Disease dataset images	13,324
Disease model size	~14 MB
API endpoints	40+
Crops (risk advisor)	12+
Crops (disease detection)	5
Buyer crops supported	14 (bilingual)
Weather forecast range	16 days
Historical weather	10 years
Confidence intervals	80%, 90%, 95%
Total development cost	Rs.0 (open-source/free-tier)

11. Testing & Validation

11.1 ML Model Validation

- Crop Risk: 5-fold cross-validation, focal-loss tuned, stratified split
- Price Model: Walk-forward validation (temporal split), conformal calibration on held-out set
- Disease CNN: 80/20 train-val split, fine-tuning from ImageNet weights, per-class accuracy logged

11.2 API Testing

- All 40+ endpoints tested via FastAPI /docs (Swagger UI)
- Auth flow tested: signup -> login -> JWT token -> protected endpoints
- Error handling validated: malformed inputs, unauthorized access, missing data

11.3 Frontend Testing

- Responsive UI tested on mobile (360px) to desktop (1440px)
- APK tested on physical Android device (Samsung, Xiaomi)
- On-device TF.js inference tested in Capacitor WebView

12. Future Scope

- Expand to all Indian states (currently Maharashtra only)
- Add multi-language support (Hindi, Kannada, Telugu)
- Integrate satellite imagery (NDVI) for field-level crop monitoring
- SMS-based alerts for price changes and weather warnings
- Implement ViT (Vision Transformer) for disease detection with larger dataset
- Add offline-first capability with service workers for low-connectivity areas
- Government subsidy information integration

13. Conclusion

MANDIMITRA successfully demonstrates that AI and ML can be applied to real agricultural problems at production scale. The system processes 3.5M+ government records, delivers 93.47% disease detection accuracy, and provides actionable price forecasts with R2=0.933. The Marathi-first approach ensures accessibility for the target user base. The three-model architecture (Risk + Price + Disease) provides comprehensive decision support covering the entire crop lifecycle -- from pre-sowing risk assessment to post-harvest price optimization. With zero development cost using open-source tools and free-tier cloud services, the project proves that impactful AgriTech solutions can be built and deployed affordably.

14. References

- Sandler, M. et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR 2018.
- Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 2017.
- MT-CYP-Net (2025). Multi-task Crop Yield Prediction. arXiv:2505.12069.
- NeuralCrop (2025). Physics-informed Neural Features for Crop Modeling. arXiv:2512.20177.
- Shafer, G. & Vovk, V. (2005). A Tutorial on Conformal Prediction. JMLR.
- AGMARKNET - Data.gov.in. Agricultural Marketing Information Network.
- NASA POWER - Prediction of Worldwide Energy Resources. https://power.larc.nasa.gov/
- Open-Meteo - Free Weather API. https://open-meteo.com/
- IRRI Rice Knowledge Bank - International Rice Research Institute.
- ICAR - Indian Council of Agricultural Research.
- FAO - Food and Agriculture Organization Plant Health Guidelines.